

NILAK-Nyanza

Faculty: CIS

Academic Year: 2025/2026

Module: Web Technology

Level of the study: III/Weekend

Category of Assessment: Final Project

Maximum: 40 marks

Due Date: 29/11/2025

Lecturer: NGIRIMANA Schadrack

Project Title

Banka—Lightweight Core Banking Web Application (HTML, CSS, JavaScript)

The system has **three main roles**:

- **Client**
- **Staff**
- **Admin**

All data must be persisted using **either**:

- **LocalStorage**
- **File-based storage (example: JSON, etc.)**
- **backend API** (NodeJS, NestJS, Django, FastAPI, Flask, Spring Boot), but PHP is seriously prohibited

You are free to choose **one** storage approach, but it must be clearly implemented and consistent.

1. Features & Modules (40 marks)

A. User Authentication (10 marks)

1. Sign Up (2 marks)

- Users register with:
 - Email address
 - Password
 - First name
 - Last name
- Save user records in your chosen storage system.

2. Email Verification with OTP (2 marks)

- After registration, generate an **OTP code** (e.g., 4 or 6 digits).
- This OTP should be sent to the user's email
- The user must enter the correct OTP to be **marked as verified** in storage.

3. Sign In (2 marks)

- Only **verified** users can log in.
- Authentication uses **email and password**.
- Maintain user session using `localStorage/sessionStorage`

4. Forgot Password (2 marks)

- The user enters their email.
- The system generates an **OTP** and sends it to users and validates it.
- If the OTP is correct, the user can enter a **new password**, which is saved in storage.

5. Admin Creates Staff/Admin Accounts (2 marks)

- Admin can create:
 - Staff users with **role = "staff"**
 - Other admins with **role = "admin"**
- Created users must be able to log in like normal.

Client (6 marks)

- The user (client) can create an account.
- The user (client) can view the account transaction history.
- The user (client) can view a specific account transaction.

Cashier (4 Marks)

- Staff (cashiers) can debit the user's (client's) account.
- Staff (cashiers) can credit the user's (client's) account.

Admin & Cashier (4 marks)

- Admin/staff can view all user accounts.
- Admin/staff can view a specific user account.

Admin (10 marks)

- Admin can activate or deactivate an account.
- Admin can delete a specific user account.
- Admin can activate or deactivate users
- Admin can delete users
- Admin can get the dashboard showing all accounts, dormant accounts, and active

accounts, money deposited and withdrawn, using a date range

UI & Extra Features (6 marks)

1. Clean UI (2.5 marks)

- Build a clear and user-friendly interface using **HTML, CSS, and JS**:
 - Appropriate Navigation bar
 - Separate views for:
 - Passenger (user dashboard, trip list, bookings)
 - Admin (trip management, stats)
 - Consistent color scheme, spacing, and typography.
- You are not allowed to use CSS frameworks (**Bootstrap, Tailwind, etc.**), just pure CSS.

2. Your Own 2 Features (2.5 marks)

Add at least **two additional features** not in the main requirements.

Explain these features in your **README (1 Mark)**